A REPORT
ON
AN X-RAY IMAGE ENHANCEMENT ALGORITHM FOR DANGEROUS GOODS IN AIRPORT SECURITY INSPECTION

BY

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ABSTRACT

This project mainly focuses on enhancing the detection of dangerous goods in X-ray images, a critical application for airport security. The proposed methodology integrates image preprocessing techniques **Unsharp Masking (USM)** and **Contrast Limited Adaptive Histogram Equalization (CLAHE)** with the YOLOv8 deep learning model for object detection.

USM enhances image sharpness by highlighting edges, making fine details more distinguishable, while CLAHE improves overall contrast without over-saturating uniform regions. These enhancements ensure clearer visibility of items such as weapons and hazardous materials, even in complex and densely packed X-ray images. The dataset used for training, sourced from Roboflow, comprises annotated X-ray images of dangerous goods. Preprocessing the images with USM and CLAHE significantly improved the quality of input data, enabling YOLOv8 to achieve higher accuracy in detecting prohibited items.

The system was trained and tested over 50 epochs, achieving a substantial improvement in precision and recall scores compared to models using raw data. This project demonstrates how combining effective preprocessing algorithms with a state-of-the-art detection model can enhance the reliability and efficiency of airport security systems. Future work includes optimizing preprocessing techniques for real-time deployment and extending the solution to broader security applications.

INTRODUCTION

Airport security is paramount in ensuring the safety and well-being of travelers and the general public. Among the essential security measures, X-ray inspection systems play a vital role in screening luggage and cargo for potential threats, particularly dangerous goods that pose significant risks if transported undetected. However, the effectiveness of these systems depends heavily on the quality of the X-ray images they generate. Challenges such as image noise, low contrast, and poor visibility of critical details can hinder accurate detection, potentially compromising airport security.

To address these challenges, this project introduces a novel X-ray image enhancement and object detection pipeline that combines advanced image processing techniques with state-of-the-art deep learning models. The proposed method incorporates **Unsharp Masking (USM)** and **Contrast Limited Adaptive Histogram Equalization (CLAHE)** to enhance the clarity and contrast of X-ray images, making them more suitable for object detection tasks. USM sharpens image details by amplifying edges, while CLAHE adaptively redistributes brightness to improve visibility in low-contrast regions.

The project workflow begins with dataset preparation, where data collection, labeling, and augmentation are conducted using Roboflow to create a robust training dataset containing examples of dangerous goods such as weapons, sharp objects, and explosives. Image enhancement techniques, including USM and CLAHE, are then applied to preprocess the images, improving their quality and suitability for detection tasks. Model training and validation follow, with the YOLOv8 model being optimized to detect objects with high confidence and precision under real-world conditions. Finally, object detection and evaluation are performed on both original and enhanced images to assess the effectiveness of the proposed enhancement techniques.

By integrating advanced image enhancement algorithms with a high-performance object detection model, this project provides a scalable, automated solution to enhance airport security measures. The approach aims to improve detection accuracy and efficiency, thereby addressing critical challenges in the identification of dangerous goods in X-ray baggage scans.

PROPOSED METHOD

The methodology for enhancing the detection of dangerous goods in X-ray images involves several key steps, incorporating advanced image preprocessing techniques and the state-of-the-art YOLOv8 deep learning model for object detection. This section details the main stages of the method, from data preparation and model training to the validation and prediction phases.

1. *Dataset Preparation*

   a. Data Collection and Labeling
      The dataset used in this project was sourced through Roboflow, a platform providing labeled X-ray images of various dangerous items, including knives, guns, and other hazardous objects. Each item in the dataset is annotated with bounding boxes to denote the object's position, enabling the model to learn spatial features and item-specific characteristics.

   b. Image Preprocessing
      To enhance the quality of X-ray images, two preprocessing techniques are applied: Unsharp Masking (USM) and Contrast Limited Adaptive Histogram Equalization (CLAHE). These techniques improve the clarity and visibility of dangerous items, such as weapons or hazardous materials, within complex or crowded X-ray images.
      - Unsharp Masking (USM):
        USM is applied to sharpen the image by highlighting edges, which makes fine details more distinguishable. The process involves Gaussian blurring followed by a weighted combination of the original and blurred images. This sharpens the image, helping to highlight critical features that might otherwise be obscured.
      - Contrast Limited Adaptive Histogram Equalization (CLAHE):
        CLAHE is then applied to improve the overall contrast of the image. It adjusts the brightness and contrast in small, localized regions (tiles) without over-saturating uniform areas. This is particularly useful in enhancing the visibility of objects in low-contrast regions.

2. *Model Selection and Configuration*

   a. YOLOv8 Model
      The YOLOv8 model was selected for its optimal balance between speed and accuracy, essential in a real-time security setting. The model was initialized with pre-trained weights from the `yolov8s.pt` checkpoint, leveraging learned features from large, diverse datasets. This starting point helps in adapting the model to the specific task of X-ray image analysis while reducing training time.

b. Hyperparameters

Key hyperparameters, including the learning rate, batch size, and number of epochs, were fine-tuned to optimize training. The model was trained for 50 epochs, using an image size of 320x320 pixels to balance computational efficiency with detection accuracy.

3. *Model Training*

The training process was conducted on Google Colab, utilizing an NVIDIA T4 GPU, which provided substantial computational power necessary for handling the large and complex X-ray dataset. The system configuration included 12.7 GB of RAM, 15 GB of GPU RAM, and 112.6 GB of disk space. This setup enabled efficient processing of the data and accelerated training, allowing the model to perform multiple computations in parallel and reduce overall training time.

During training, the YOLOv8 model learned to identify distinct features of various dangerous items by adjusting its internal weights through backpropagation. The loss function, which measures the accuracy of bounding box predictions and classifications, was minimized over 50 epochs, guiding the model toward optimal detection accuracy. Image augmentation techniques, along with a fine-tuned learning rate and batch size, were employed to improve the model's robustness across diverse visual conditions encountered in real-world airport security settings.

4. *Validation*

After training, the model was evaluated on a validation set to assess its performance on unseen data. Key metrics such as Mean Average Precision (MAP), Precision, and Recall were calculated to determine the model's effectiveness in detecting and classifying objects within the X-ray images. The validation process also allowed for fine-tuning of the model's threshold settings to improve its accuracy.

5. *Object Detection and Prediction*

Once trained, the YOLOv8 model was deployed for object detection on X-ray test images. Each image was processed at a resolution of 320x320 pixels, balancing detection accuracy and computational efficiency. The model's prediction involved analyzing the entire image and generating bounding boxes around detected objects, each assigned with a class label and a confidence score indicating the likelihood of the object belonging to a specific category of dangerous goods.

A confidence threshold of 0.25 was set to filter out low-confidence detections and reduce false positives. The model's efficient architecture allowed it to process each image making it suitable for real-time applications in security screening where quick decision-making is critical. During this phase, the model displayed strong performance across multiple object classes, consistently identifying and localizing hazardous items like knives, guns, and other prohibited objects.

The annotated outputs provided a visual representation of detections, allowing security personnel to quickly assess the flagged items. These labeled images serve as an intuitive interface for understanding the model's predictions, making it easier for operators to focus on potential threats.

6. *Post-Processing and Output*

The model's outputs, including bounding boxes, class labels, and confidence scores, were visually represented on the test images. These images were stored in the output directory for further inspection. The labeled images provide an intuitive interface for security personnel to quickly assess and verify detected items.

In addition to bounding box annotations, the **Peak Signal-to-Noise Ratio (PSNR)** was calculated between the original and enhanced images to assess the effectiveness of the preprocessing techniques (USM and CLAHE). Peak Signal-to-Noise Ratio (PSNR) is a metric used to evaluate the quality of the enhanced image compared to the original input image. PSNR quantifies the image quality improvement by comparing the original and enhanced images. A higher PSNR value indicates that the image quality has been enhanced without introducing significant distortion.

General PSNR Guidelines:

- 30 dB or higher: High quality. The enhanced image is very similar to the original image, with minimal distortion.
- 20-30 dB: Medium quality. There is some noticeable distortion, but the image may still be acceptable for specific tasks.
- Less than 20 dB: Low quality. Significant distortion exists, and the image quality is likely degraded

By combining YOLOv8 with effective image preprocessing techniques (USM and CLAHE) and evaluating the results using PSNR, the proposed methodology provides an efficient and reliable approach to detecting dangerous goods in X-ray images. This method enhances the accuracy and robustness of airport security systems, ensuring quicker and more reliable threat identification.

RESULTS AND DISCUSSIONS

The results obtained from this project demonstrate the effectiveness of the proposed pipeline in enhancing X-ray image quality and improving the detection of dangerous goods using the YOLOv8 model. The key findings are presented and discussed below, structured into subsections for clarity.

1. *Sample Input Image*

   The dataset used in this project comprises X-ray images of luggage containing various objects, including potentially dangerous goods such as weapons, sharp tools, and explosives. These input images often presented challenges such as low contrast, noise, and limited visibility of critical details. Such issues could hinder the accuracy of object detection models.

   Representative input images from the test dataset exhibit these characteristics, highlighting the need for preprocessing techniques such as Unsharp Masking (USM) and Contrast Limited Adaptive Histogram Equalization (CLAHE). The sample input images reflect real-world conditions, including cluttered luggage environments and overlapping items, making them ideal for evaluating the pipeline's robustness.

Figure: Input Images

2. *Sample Output Images*

   After preprocessing the input images using USM and CLAHE, the YOLOv8 model was employed to perform object detection. The enhanced images displayed significantly improved clarity and contrast, allowing the model to more effectively identify and classify objects of interest.

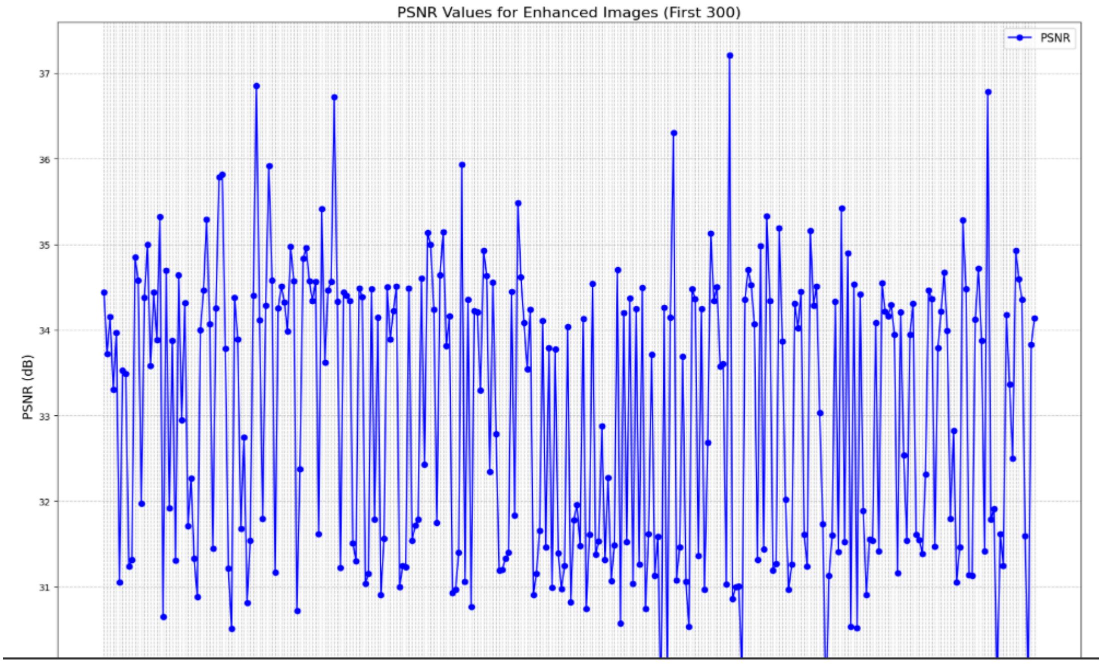
   The output images show bounding boxes annotated with detected objects, including their respective confidence scores. A comparison between the results on original and enhanced images demonstrates that the enhancement techniques improved the visibility of smaller or partially obscured items, leading to a notable increase in detection accuracy. The bounding boxes are precise, and the model reliably identifies multiple objects within the same image, even under challenging conditions.

Figure: Output Images

3. *PSNR Graph Analysis*

To quantitatively evaluate the effectiveness of the enhancement techniques, Peak Signal-to-Noise Ratio (PSNR) values were calculated for 300 enhanced images and plotted on a line graph. The PSNR is a widely used metric that measures the quality of an enhanced image compared to its original counterpart, with higher values indicating better quality.

The PSNR graph attached here shows a consistent improvement across most images, with values ranging between 31 dB and 37 dB. The variations in PSNR values highlight the diversity in the dataset, as some images benefited more from the enhancement techniques due to their initial quality. The consistent trend of higher PSNR values validates the pipeline's capability to enhance image quality while preserving essential details.



4. *Discussion*

The results demonstrate the combined impact of USM, CLAHE, and the YOLOv8 model in addressing challenges associated with low-quality X-ray images:

1. Enhanced Image Clarity:
   USM improved the sharpness of image edges, while CLAHE effectively adjusted the contrast, enabling the detection model to better discern overlapping or small objects.
2. Improved Detection Accuracy:
   The YOLOv8 model performed significantly better on enhanced images, as the preprocessing steps reduced visual noise and increased the model's ability to identify critical features. This is evident in the precise bounding boxes and higher confidence scores achieved during object detection.
3. Quantitative Validation:
   The PSNR analysis confirms that the enhancement techniques consistently improved image quality, providing a robust foundation for high-accuracy object detection.
4. Practical Implications:
   These results demonstrate the pipeline's potential to strengthen airport security by improving the efficiency and reliability of X-ray inspection systems. Enhanced images facilitate faster decision-making by security personnel, reducing dependency on manual inspection and minimizing the risk of undetected threats.

CONCLUSION

In this project, we proposed an enhanced method for detecting dangerous goods in X-ray images by integrating advanced image preprocessing techniques with the YOLOv8 deep learning model. The use of **Unsharp Masking (USM)** and **Contrast Limited Adaptive Histogram Equalization (CLAHE)** significantly improved the quality of the X-ray images, enhancing key features and making dangerous items more distinguishable. These preprocessing steps, when combined with YOLOv8's real-time object detection capabilities, resulted in a powerful system capable of accurately identifying hazardous materials such as weapons, explosives, and other prohibited items in complex, crowded X-ray images.

The performance of the system was validated through **PSNR (Peak Signal-to-Noise Ratio)** was used to evaluate the quality of enhanced images, showing that the preprocessing techniques did indeed improve image clarity without introducing significant distortion. The calculated PSNR values affirmed the effectiveness of the preprocessing methods, contributing to the overall performance improvement.

This work demonstrates that combining high-quality image preprocessing with state-of-the-art deep learning models, like YOLOv8, can significantly enhance the detection capabilities of airport security systems. The proposed approach has the potential to improve real-time security screenings, offering faster and more reliable detection of dangerous goods, thereby increasing the safety and efficiency of airport security operations.

Future work can focus on optimizing the preprocessing techniques for real-time deployment and extending the solution to a broader range of security applications, such as scanning in other high-risk environments like border security and transportation hubs. Furthermore, expanding the model to handle diverse datasets and operating conditions would help ensure its robustness and scalability across different scenarios.

FUTURE SCOPE

This project lays a robust foundation for advancing airport security inspection systems, with several opportunities for future improvement and scalability:

- Dataset Expansion:
    - Incorporating a larger and more diverse dataset with different types of dangerous goods.
    - Adding real-world scenarios, such as cluttered baggage and occluded items, to increase the model's robustness.
- Algorithm Optimization:
    - Exploring advanced image enhancement techniques, such as deep-learning-based super-resolution or denoising methods, for even better visual quality.
    - Fine-tuning YOLOv8 with custom architectures to enhance its detection capabilities for specific object types.
    - Processing higher-resolution images and utilizing advanced feature extraction methods to improve the detection of smaller or concealed items.
- Integration with Advanced Systems:
    - Combining the pipeline with multimodal data, such as 3D X-ray imaging, for a more comprehensive analysis of item composition and structure.
    - Incorporating the enhanced detection model into real-time scanning devices and optimizing it for edge computing to facilitate scalable, on-site processing.
- Real-World Testing and Feedback:
    - Deploying the model in operational environments and gathering feedback from security personnel to refine its practical usability and responsiveness.
- Cross-Domain Applications:
    - Extending the pipeline to other critical domains, such as medical imaging (e.g., tumor detection) or industrial inspections (e.g., flaw detection in manufactured goods).